MIL-N-36319A
9 April 1970
SUPERSEDING
MIL-N-36319
2 November 1964

MILITARY SPECIFICATION

NEEDLES, HYPODERMIC, CARTRIDGE TYPE, DISPOSABLE

This specification is mandatory for use by all the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, disposable, cartridge type, hypodermic needles.

1.2 Classification.

1.2.1 Types. Needles shall be of the following types, as specified (see 6.1):

Type I. Plastic hub (self-threading or threaded).
Type II. Aluminum hub (threaded).

1.2.2 Sizes. Needles shall be of the following sizes (gages and lengths) as specified (see 6.1):

Size 1. - 25 gage, 13/16 inch.
Size 2. - 25 gage, 1-3/8 inch.
Size 3. - 27 gage, 1-3/8 inch.
Size 4. - 27 gage, 13/16 inch.

2. APPLICABLE DOCUMENTS

FSC 6515

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2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-S-36586	Sterilization Strip Test, Bacterial Spore

STANDARDS

Federal

FED-STD-151	Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Medical Supply Center

20467	Needle, Cartridge.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals, shall apply:

American Iron and Steel Institute

Steel Products Manual, Stainless and Heat-Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, N. Y. 10017.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.(U.S.P.).

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

3. REQUIREMENTS

3.1 Material.

3.1.1 Cannula. The cannula shall be fabricated from either seamless drawn, or welded and drawn, stainless steel tubing, American Iron and Steel Institute (AISI) type 304 stainless steel. Material shall be tested, as specified in 4.4.6, for chemical composition.

3.1.2 Hubs. The material for the Type I hub shall be polypropylene plastic. The material for the Type II hub shall be anodized or chemically treated aluminum alloy.

3.1.3 Needle container. The needle container shall be fabricated of polypropylene or other suitable plastic, and shall maintain sterility of contents (see 3.8 and 3.13).

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3.1.4 Insert. Insert, if present, shall be fabricated of an aluminum alloy, suitable for the purpose intended.

* 3.2 Style, design, and construction. Shall be a sterile cartridge type hypodermic needle, packaged in an individual sealed container. Needle shall be fabricated of material as specified in 3.1.1, 3.1.2, and 3.1.4. The needle container shall consist of a sheath and cap which shall be fabricated of material as specified in 3.1.3. The needles shall be suitable for use with 6515-985-7106, SYRINGE, CARTRIDGE, Aspirating, Thumb Ring Handle: Dental and 6515-559-3000, SYRINGE, CARTRIDGE, Crutch Handle: Dental. Type I or type II needle and its individual container shall be in accordance with Drawing 20467.

3.3 Corrosion resistance.

3.3.1 Cannula. When tested as specified in 4.4.7, the cannula shall show no evidence of corrosion.

* 3.3.2 Aluminum hub, Type II. When tested as directed in 4.4.9, the aluminum hub shall show no evidence of corrosion.

3.4 Instructions. Each unit box and bag, i.e., box plus bag containing 250 needles in their individual containers, shall contain instructions to the effect that needle is disposable, is for one time use only, and shall then be discarded. The instructions shall include illustrations showing "screw-on" and "screw-in" (type I) needles and instructions for use of type II needles for use with dental syringes.

3.5 Stiffness. When tested as specified in 4.4.1, the maximum deflection of the cannula shall not exceed the distance indicated below for appropriate gage:

Gage	Maximum deflection (inch)
27	0.022
25	0.026

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3.6 Security of staking. When tested as specified in 4.4.2, the cannula or aluminum insert shall not loosen or pull out of the hub.

3.7 Resistance to breakage. When tested as specified in 4.4.3, there shall be no breakage.

3.8 Sterility. When tested as specified in 4.4.4, needles shall comply with sterility test requirements of The Pharmacopeia of the United States (U.S.P.).

3.9 Freedom from pyrogens. When tested as specified in 4.4.5, needles shall comply with the pyrogen test requirements of the U.S.P. for Transfusion and Infusion assemblies.

* 3.10 Safety (non-toxicity). When tested as directed in 4.4.6, needles shall comply with the requirements of the USP for Transfusion and Infusion Assemblies.

* 3.11 Certificates of sterility, freedom from pyrogens and safety. A certificate listing all lot control numbers shall state that representative samples from each lot or control have been tested and found to be sterile, free from pyrogens, and safe (non-toxic). This certificate shall include the specific tests performed and the test results. For destination inspection only, certificates of sterility, freedom from pyrogens, and safety shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Materiel, 2800 S. 20th St., Philadelphia, Pa. 19101, Attn: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

* 3.12 Identification marking. Each individual needle container shall be marked with the name of the manufacturer, or registered trademark, gage, and needle length, i.e., short or long, and in addition, shall be marked "Dispose after one use," or similar legend.

3.13 Closure of needle container. The needle container shall be so closed that it shall be tamperproof (i.e., an opened container shall be readily discernible and cause for rejection).

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3.14 Workmanship. Workmanship shall be first class throughout. Needles shall be free from defects which detract from their appearance or may impair their serviceability. The cannula shall show a smooth finished outer surface free from pits and tool marks, and inner surface free from scale, deep pits, deep furrows, or die marks; the lumen of the cannula shall be free from chips, dirt, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the stainless steel and plastic may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

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* 4.2.1 For examination. Sampling for nondestructive examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination purposes shall be one hypodermic needle.

Table I. Sampling for Examination

Type of Defect	Inspection Level	AQL (Percent defective)*
Major	I	1.0
Minor	I	2.0

* If acceptance number is not exceeded, lot shall be acceptable. If rejection number is equalled or exceeded, inspection at level II may be performed to determine final acceptance or rejection.

* 4.2.2 Sampling for test. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. Where possible the same sample shall be used for the determination of two or more test characteristics.

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Table II. Sampling for Tests

Item or Component	Characteristic	Req.	Test procedure	Insp. level	AQL (Percent defective)
Cannula	Composition	3.1.1	4.2.4 & 4.4.8	S-1	*
Cannula	Corrosion resistance	3.3.1	4.4.7	S-2	*
Cannula	Stiffness	3.5	4.4.1	S-1	*
Hub-cannula	Security of staking	3.6	4.4.2	S-2	*
Hub-cannula	Resistance to breakage	3.7	4.4.3	S-2	*
Hub, Type II	Corrosion resistance	3.3.2	4.4.9	S-3	*
Unit of product	Sterility	3.8	4.4.4	S-3 **	*
Unit of product	Freedom from pyrogens	3.9	4.4.5	S-3	*
Unit of product	Safety	3.10	4.4.6	S-3	*

*Acceptance number shall be zero for all sample sizes.
**For the sterility testing a minimum of 10 units from each lot shall be tested.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the construction of the needles shall be selected for chemical analysis.

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* 4.3 Examination. The needles shall be examined to determine compliance with all requirements contained in this specification. Visual examination shall be conducted under 7X magnification and good illumination. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include but shall not be limited to defects listed in Table III.

Table III. Classification of Defects

Categories	Defects
Major	
101	Presence of foreign materials, loose particles.
102	Point is not free of burrs.
103	Cannula lumen not free of obstruction because of excessive swaging action.
104	Cannula not straight.
105	Cannula exterior not smoothly finished.
Minor	
201	Cannula or hub not free of discoloration.
202	Cannula length not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Stiffness test. The cannula shall be placed across two test edges (0.002 inch radius maximum) separated by the distance indicated in Table IV below for appropriate gage. A load as indicated below, shall be applied at the center of the span and the resulting maximum deflection of the cannula shall be measured using a gage with graduations of at least 0.001 inch. The force shall be applied through an edge with a maximum radius of 0.036 inch, and centered to within ± 0.003 inch.

Table IV. Conditions for Stiffness Test

Gage	Separation of Test edges, inch (± 0.003 inch)	Load, ounces (± 1/4 ounce)
27	0.375	20
25	0.500	25

4.4.2 Security of staking. For type I needle, a straight pull of 5 pounds shall be applied between the cannula and aluminum insert for a period of 1 minute; this procedure shall be repeated by applying the pull between the aluminum insert, if present, and the plastic hub. For type II needle, a straight pull of 5 pounds shall be applied between the needle and hub for a period of 1 minute.

* 4.4.3 Resistance to breakage. The needle hub shall be held in a fixed position and a force applied to the cannula at a point 7/16 inch from the hub to deflect the 25 gage needle 25°, and at a point 5/16 inch from the hub to deflect the 27 gage needle 30°. The radius so described for each size needle shall be maintained as the cannula is moved in a clockwise followed by a counterclockwise circle. The formation of a clockwise and a counterclockwise circle shall constitute one cycle. The needle shall withstand 20 such cycles without separation or becoming loose at the hub-cannula bond.

* 4.4.4 Sterility. Testing for sterility shall be in accordance with the method specified in the USP Procedural Details for Sterility Tests of Class B, Type II, Inoculated Products, and for Parenteral Devices. As an

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alternate, testing for sterility shall be in accordance with the method specified in the USP Procedural Details for Sterility Tests, Class B, Type II, Inoculated Carrier, and Parenteral Devices, except that the inoculated carrier shall contain sterilization test strips carrying spores of Bacillus subtilis var. globigii and Bacillus stearothermophilus, Type B of MIL-S-36586.

4.4.5 Freedom from pyrogens test. Place 10 needles in a sterile pyrogen-free flask, pour 250 milliliters (ml.) of sterile, pyrogen-free saline test solution (U. S. P.) into the flask and then heat the contents of the flask at not less than 40°Centigrade (C.) nor higher than 50° C. for 1 hour. The resulting saline extract shall be submitted to the "Pyrogen Test" as described in the current edition of the U. S. P.

* 4.4.6 Safety (non-toxicity). Safety shall be determined in accordance with the method in the USP for Transfusion and Infusion Assemblies.

4.4.7 Corrosion-resistance of cannula. The cannula (after removal from hub) shall be immersed in 10 percent citric acid at room temperature for 5 hours and, after removal from the acid, shall be boiled in distilled water for 30 minutes, and then shall remain immersed for 48 hours in the distilled water. At the end of the 48-hour period, the cannula shallbe allowed to dry by evaporation, and shall then be examined for evidence of corrosion. A glass vessel shall be used throughout this test.

4.4.8 Chemical analysis. Cannula material (see 3.1.1) shall be subjected to chemical analysis in accordance with methods 111 or 112 of Standard No. 151.

* 4.4.9 Corrosion resistance of aluminum hubs. Needles with aluminum hubs shall be placed in test tubes after the cannula has been snipped off to leave a 1/16 inch distal protrusion from the hub. A single hub shall be placed in a test tube. The open end oi the hub shall be up. The hub shall then be covered with sodium mercaptomerin (FSN 6505-126-9425) at 20° to 25° C. (68° to 77°F). Exposure shall be maintained for one hour. During this exposure there shall be no discoloration visible nor any sign of a reaction.

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5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packing shall be level A as specified (see 6.1).

* 5.1.1 Unit of issue. One (1) box containing 250 needles, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	12 units

* 5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Individual package. Each needle shall be packaged in an individual container as specified in 3.2. Closure of container shall be as specified in 3.13. The completed sealed container shall be substantially free from foreign matter and shall maintain the degree of sterility specified in 3.8 until after the seal is broken or package is opened.

* 5.1.3.2 Unit package. Two hundred and fifty individual packages shall be packaged in a polyethylene bag of appropriate size having a wall thickness of not less than 0.0025 mils. Closure shall be effected by hot sealing or by use of plastic coated wire ties. Each bag shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-636, class domestic. Method I closure shall be utilized.

* 5.2 Packing. Packing shall be level A or B as specified (see 6.1).

5.2.1 Level A.

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* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

* 5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.3 Marking.

5.3.1 Individual package. Each individual package shall be marked as specified in 3.12.

* 5.3.2 Unit package. At least one (1) side face of each unit carton and each bag shall bear instructions and illustration of the recommended technique for opening the individual package and for applying needle to the syringe (see 3.4). The method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-9/69-G". Final sterilization shall be on or after indicated packaging date. In addition, the following information shall be printed on the carton or applied by means of a label:

Federal stock number.
Item identification.
Unit and quantity.
Name (s) of manufacturer (and contractor).
Lot (control) number.
Legend indicating that needles are sterile, pyrogen-free, and disposable type.
Contract number.
Level of preservation and date.

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* 5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) numbers shall be shown. Marking shall include a legend indicating that needles are sterile, pyrogen-free, and disposable type. The method of sterilization as specified in 5.3.2 shall be shown.

NOTE: The item identification shall be shown on interior packages and exterior container, as follows:

Item identification: Needle, Hypodermic, Cartridge Type, Disposable, *Gage, *inch, 250s.

*Applicable information to be inserted.

6. NOTES:

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Type required (see 1.2.1).

(d) Selection of applicable levels of packaging and packing packing (see 5.1 and 5.2).

* 6.2 This specification is applicable to the following Federal Supply Catalog items:

6515-817-2275	NEEDLE, HYPODERMIC, Cartridge Type, Disposable, 25 Gage, 13/16 inch, 250s.
6515-817-2276	NEEDLE, HYPODERMIC, Cartridge Type, Disposable, 25 Gage, 1-3/8 inch, 250s.
6515-817-2277	NEEDLE, HYPODERMIC, Cartridge Type, Disposable, 27 Gage, 1-3/8 inch, 250s.
6515-817-2278	NEEDLE, HYPODERMIC, Cartridge Type, Disposable 27 Gage, 13/16 inch, 250s.

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6.3 This specification is intended to cover the types which are normally procured to meet Military requirements. It does not include all classes, types, or sizes as indicated by the title nor all types which are commercially available.

* 6.4 The margins of this speci ication are marked with an asterisk to indicate where changes (additions, modi ications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM.

Project No. 6515-1465

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

MIL-M-36319A FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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